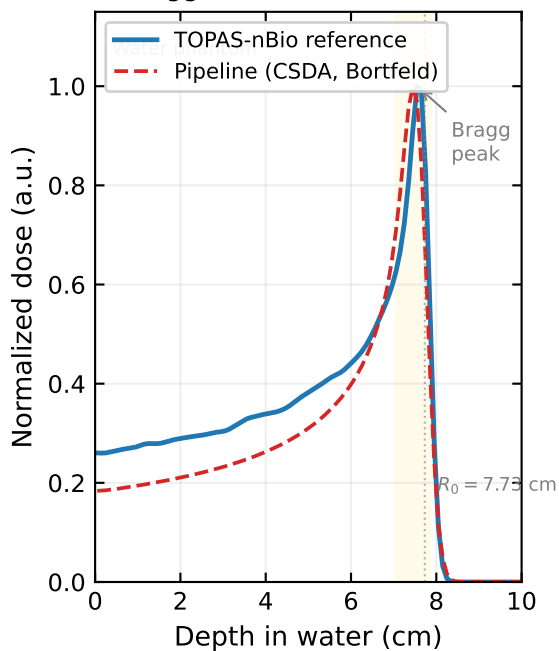
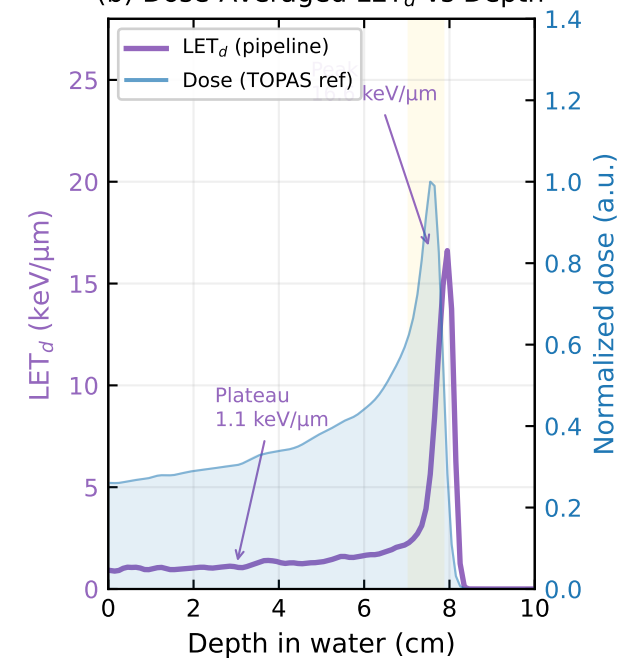


TOPAS-nBio Benchmark: 100 MeV Proton Beam in Water Phantom

(a) Bragg Curve — 100 MeV Proton



(b) Dose-Averaged LET_d vs Depth



(c) Proton RBE vs Depth — Pipeline vs Clinical Models (100 MeV beam, $D_{ref} = 4$ Gy)

